Mathematics Lecture Notes

June 3, 2026

Contents

Chapter 1

Introduction

These notes collect selected mathematical ideas that support precise reasoning, modelling, and problem solving. The course begins with linear algebra, where vectors, matrices, and systems of equations provide a language for describing structure. It then moves to analytic geometry, where algebraic equations are connected with geometric objects, and finally to calculus, where functions, limits, continuity, and derivatives describe change.

The purpose of the notes is not only to list formulas, but also to develop a habit of reading mathematical definitions carefully and using notation consistently. Each topic should be treated as part of a larger toolkit: definitions introduce objects, examples show how they behave, and calculations illustrate how abstract concepts become useful in concrete problems.

Chapter 2

Linear Algebra

This chapter introduces the basic objects and methods of linear algebra.

2.1 Vectors

This section will introduce vectors and their basic operations.

2.2 Matrices

This section will introduce matrices and basic matrix operations.

2.3 Systems of Linear Equations

This section will introduce systems of linear equations and their connection with matrices.

Chapter 3

Analytic Geometry

This chapter introduces geometric objects described using coordinates and algebraic equations.

3.1 Coordinates

This section will introduce coordinate systems and the representation of points.

3.2 Lines

This section will introduce equations of lines and their geometric interpretation.

3.3 Planes

This section will introduce equations of planes in three-dimensional space.

Chapter 4

Calculus

This chapter introduces functions, limits, continuity, and derivatives.

4.1 Functions

This section will introduce functions as mathematical objects describing dependence between quantities.

4.2 Limits

This section will introduce the concept of a limit.

4.3 Continuity

This section will introduce continuity and its basic interpretation.

4.4 Derivatives

This section will introduce derivatives as rates of change.